WOOSEOP HWANG

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Education

University of Oxford Oct 2023 – Jun 2024
Master of Mathematics and Theoretical Physics (MMathPhys) *Oxfordshire, United Kingdom*
- First Class Honours.

University of Oxford Oct 2020 – Jun 2023
Bachelor of Arts in Physics (BA) *Oxfordshire, United Kingdom*
- First Class Honours, with Distinction.

Experience

Qraft Technologies Aug 2024 – Present
AI/Quant Researcher (Alternative Military Service) *Seoul, South Korea*
- Developed and managed live high-frequency ETF arbitrage strategies in production for Korean markets, spanning short-horizon signal modelling through to low-latency execution.
- Developed reinforcement-learning–based order execution algorithms, designing reward shaping and state representations to optimize market impact under partial observability.
- Built end-to-end systematic trading algorithms bridging deep RL research and production deployment, with emphasis on sample efficiency and robustness to regime shifts.

data cybernetics ssc GmbH Oct 2023 – Mar 2024
Researcher *Remote, Germany*
- Investigated applications of quantum state preparation, including Quantum Convolutional Neural Networks, portfolio optimization, and the solution of linear systems via the HHL algorithm.
- Evaluated the effectiveness of quantum-inspired classifiers on a fingerprint recognition task.
- Developed a Python library to transpile Qiskit quantum circuits into their PennyLane equivalents.

Yonsei University Dec 2022 – Aug 2023
Student Researcher *Seoul, South Korea*
- Developed an efficient classical simulation of the Helstrom measurement for an arbitrary number of quantum state copies.
- Investigated contextuality in quantum circuits.

Publications

Preparing Ground and Excited States Using Adiabatic CoVaR 2024 – 2025
First-author, with Prof. Bálint Koczor (University of Oxford) *Oxford, UK*
- Developed a hybrid quantum-classical algorithm for preparing ground and excited eigenstates of target Hamiltonians, addressing the warm-start limitation of CoVaR via adiabatic Hamiltonian morphing. Published in *New Journal of Physics* (2025).

Generalized Gaussian Temporal Difference Error for Uncertainty-aware Reinforcement Learning 2024
Co-author, with S. Kim, J. Lee, N. Cho, S. Han (Qraft Technologies) *Seoul, South Korea*

- Proposed a generalized-Gaussian model of temporal-difference error capturing higher-order moments (kurtosis), improving the estimation of aleatoric and epistemic uncertainty in deep reinforcement learning. arXiv:2408.02295.

Quantum-Inspired Classification via Efficient Simulation of Helstrom Measurement 2023 – 2024

First-author, with Prof. Daniel K. Park (Yonsei), I. F. Araujo, C. Blank *Seoul, South Korea*

- Derived an efficient simulation of the Helstrom Quantum Centroid classifier for arbitrary data-copy counts, revealing that copy number is a hyperparameter rather than a monotonic improvement and enabling datasets larger than the underlying Hilbert space dimension. arXiv:2403.15308.

Awards & Honors

Best Student Poster Award, Asia Quantum Information Science Conference (AQIS)	2024
Fitzgerald Prize, Exeter College, University of Oxford	2021, 2024
East Scholarship, Exeter College, University of Oxford	2020–2024
First Place, South East Asia Mathematics Competition	2020

Talks & Presentations

Poster presentations: Asia Quantum Information Science Conference (AQIS); Quantum Techniques in Machine Learning (QTML) 2024

Skills

Programming: Python (Advanced), C++ (Intermediate), Rust (Intermediate), LaTeX
Areas: Quantitative Research, Machine Learning / Reinforcement Learning, Quantum Computing
Languages: English (Fluent), Korean (Native)